

Project Presented by:

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LICENSE PLATE DETECTION

PROBLEM STATEMENT



- PROBLEM:
 - Manual plate logging during security rounds is slow,, and tedious
- Stakeholders:
 - Commercial property managers and security teams
- Impact:
 - Wastes patrol time, introduces human error, and delays spotting unauthorized vehicles
- Solution
 - Automated License Plate Detection for digital database cross-checks.

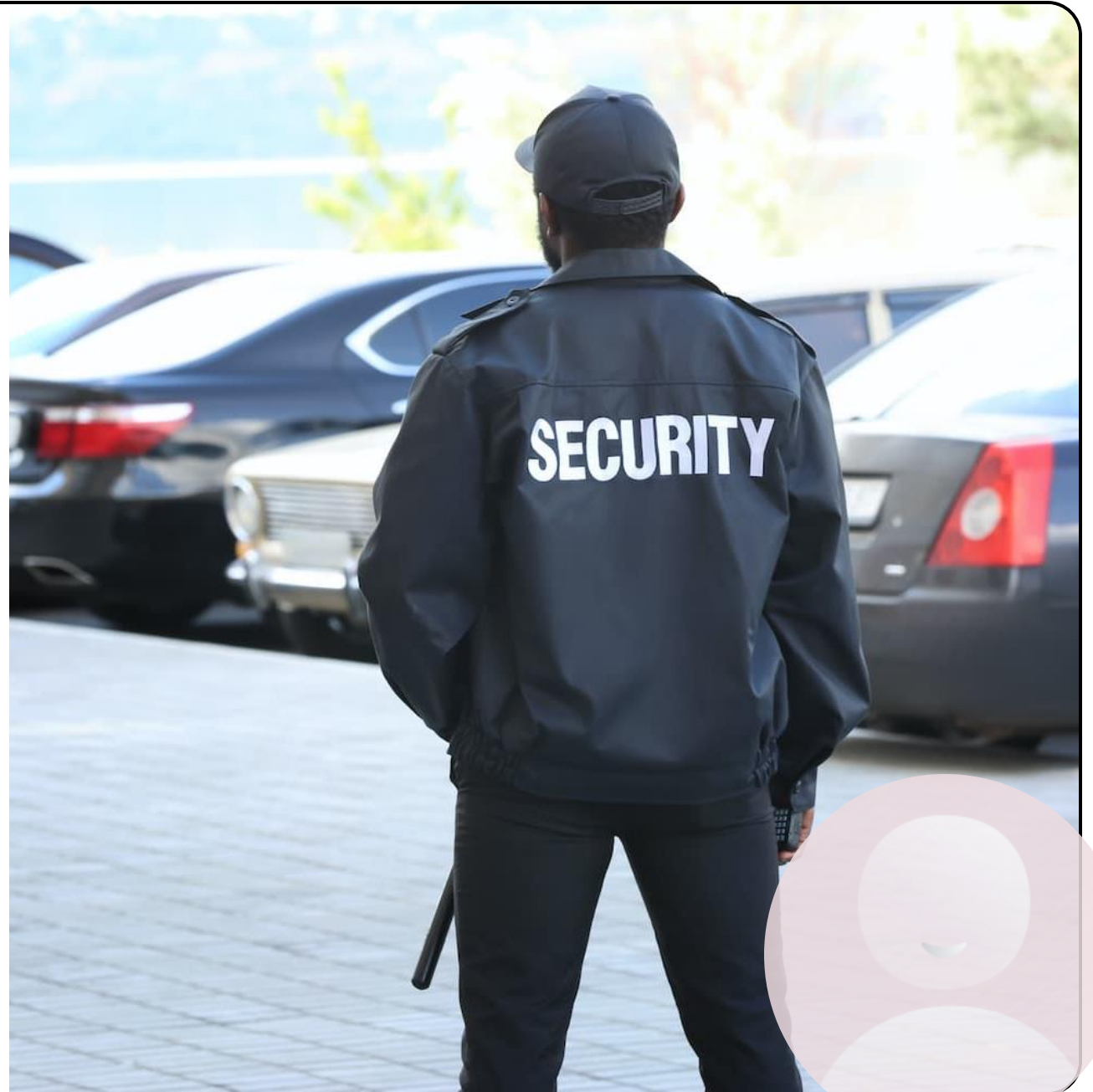
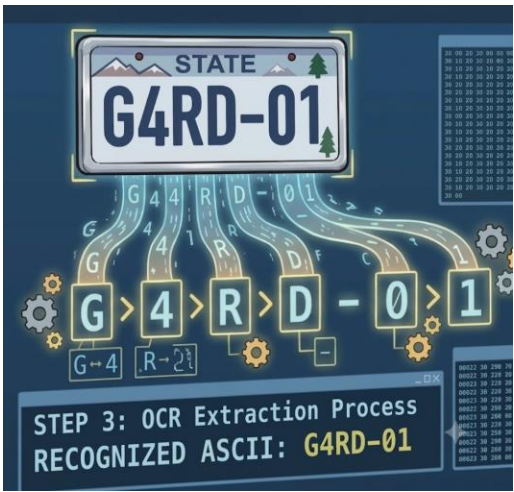




PLATE DETECT

Successful introduction
of our innovative
product to the market



DB MATCH

Database Cross-Check
& Alert: final output &
actionable intelligence



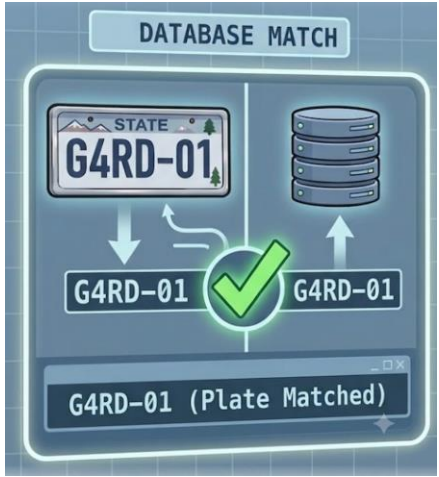
CAMERA INPUT

Successful completion
and finalization of our
innovative product



OCR EXTRACT

Converts License Plate
to Text w/EasyOCR:
Tier 2 upgrade



PRODUCT ROADMAP



COMPREHENDING MARKET TRENDS

85%

of businesses say digital transformation is essential to
their survival

Risks and Resources



Risk	Probability	Mitigation Plan
Model performance drops under varied lighting or viewing angles	<i>Low</i>	Apply image augmentation (brightness, rotation, blur) or retrain using a more diverse dataset.
EasyOCR misreads license plate characters due to low-resolution or blurred crops	<i>High</i>	Preprocess cropped plates using upscaling, grayscale conversion, and contrast enhancement before OCR.

THRIVING IN UNCERTAINTY

Adapting to change

Businesses that embrace agility respond more quickly and effectively to market shifts, customer demands, and unforeseen challenges.

Real-world impact

Industry leaders who stay ahead of disruption are those who proactively embrace change and continuously innovate.





WHERE DO WE GO
FROM HERE?

Technology Stack Overview

LAYER	TECHNOLOGY	PURPOSE IN PROJECT
 Programming	Python	Core language for AI and computer vision development.
 Environment	Google Colab	Free GPU-accelerated cloud environment for training.
 Object Detection	Ultralytics YOLO	Detects plates and generates high-precision bounding boxes.
 Image Processing	OpenCV	Loading images, preprocessing, and visual annotation.
 Dataset Source	Roboflow / Kaggle	Provides labeled license plate images for model training.

Data Acquisition Strategy



Source & Provenance

Public dataset from Roboflow Universe (Project: *rxg4e*, Version 13).



Scale & Volume

Total of 10,125 high-quality images (70% Train, 20% Valid, 10% Test).



Label Specification

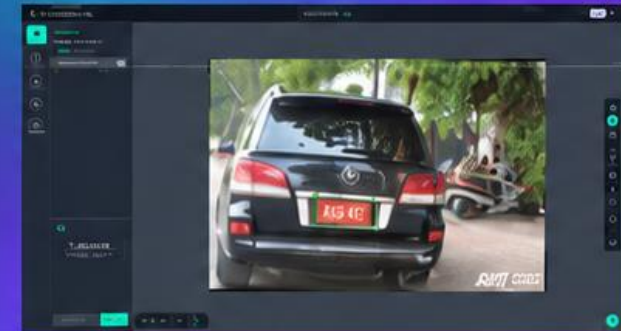
Annotated with precise bounding boxes for the class: **license-plate**.



Preparation Status

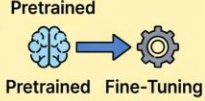
Pre-labeled and cleaned; ready for immediate YOLO training and deployment.

How To Train and Deploy an ANPR System



roboflow

CV Technical Approach

	 Required Item	 Selection	 Explanation
1	 CV Technique	 Object Detection	The model locates license plates within parking lot images by drawing bounding boxes around detected plates. 
2	 Model Architecture	 YOLO11n (You Only Look Once - Nano)	A lightweight, single-stage detection architecture designed for fast and efficient image analysis. 
3	 Model Usage	 Transfer Learning (Fine-Tuning)	Starts with a pretrained YOLO model and adapts it to recognize license plate using a labeled dataset. 
4	 Framework	 PyTorch + Ultralytics YOLO	PyTorch provides the deep learning foundation, while Ultralytics provides the YOLO implementation for training and inference. 

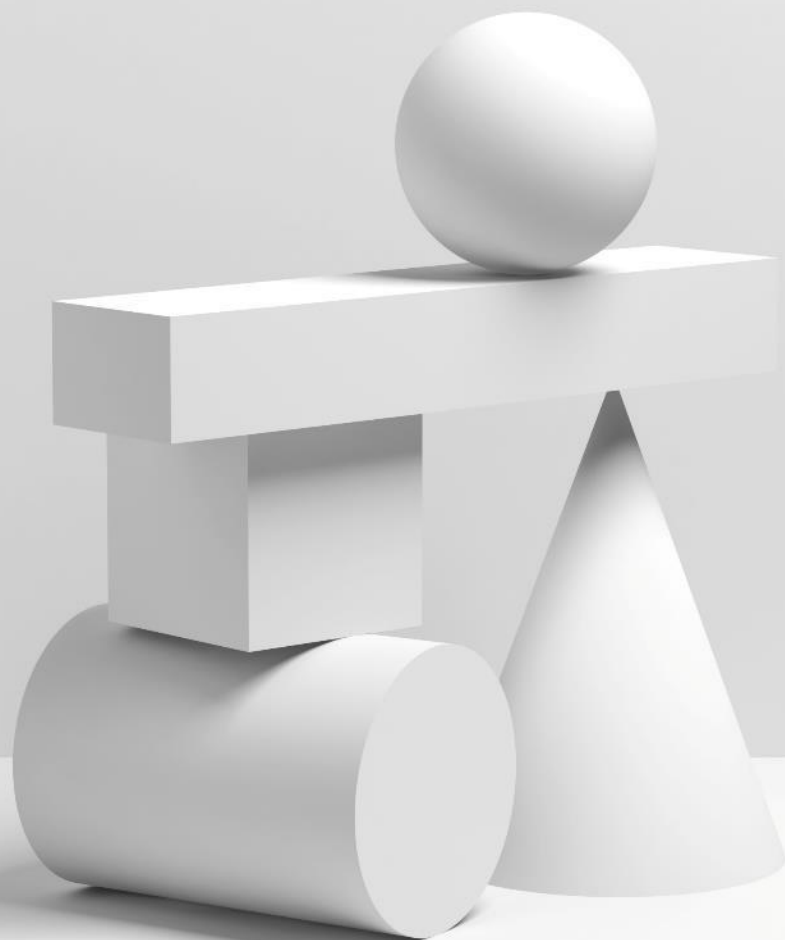
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- Yadi yadi yada

Phase	Goal	Milestone	10-week Term
BluePrint -- LPR	Plan approval	Midterm submission (plan)	07/13
First Working Demo	Proof of Concept for LPR work	Minimally Viable Product (DEV)	07/17
Make It Yours	Add My Parking Lot Data	QA on Functional Project: UAT/SIT	07/22
Improve/Measure	Test/refine Final project	Metrics Recorded (PROD)	07/27
Package & Present	Demo, README.md, slides	Final Submission for business analysis (hypercarE)	07/29

MILESTONE PLAN—License Plate Recognition CV Project

EMBRACE CHANGE. LEAD THE FUTURE.

The world is shifting and those who adapt and innovate will be the ones to lead the way.



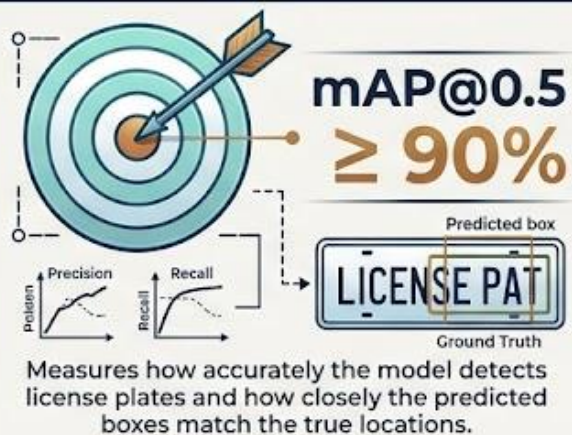
Q & A

Panel discussion session

LICENSE PLATE DETECTION (YOLOv11)

PROJECT SUCCESS METRICS

PRIMARY METRIC



SECONDARY METRIC



EVALUATION STEPS

Stage 1

- Train on labeled training images



Stage 2

- Measure mAP on labeled test images



Stage 3

- Demonstrate detections on new images from a parking lot



Data sources

Future extensions: OCR (EasyOCR)

Success Metrics